

PRINCE SINHA

✉ princesinha24august@gmail.com | 📞 9075736615 | Bengaluru, India | github.com/princesinha24 | <https://www.linkedin.com/in/prince-sinha-45689919b>

SUMMARY

Backend and Systems Software Engineer with 4 years of experience designing, developing, and optimizing scalable backend and system-level solutions, with strong problem-solving skills.

TECHNICAL SKILLS

Core Concepts: Data Structures and Algorithms, OOP, System Design, Database Management, OS

Programming Languages: Java, C/C++, Python, Perl, JavaScript

Frameworks & Technologies: Microservices, Linux, MongoDB, Node.js, React.js, Spring Boot, SQL, AWS, Model Context Protocol (MCP)

WORK EXPERIENCE

Software Engineer II, Cisco Systems, Bengaluru, India Aug 2022 – Present

- Built **microservices** for secure package upload, verification, SFTunnel-based delivery, pre-check validations, seamless upgrades with rollback/revert support, and real-time status monitoring. Incorporated a **telemetry microservice** to generate upgrade reports.
- Designed a **hitless upgrade framework** to ensure zero traffic disruption by implementing **High Availability (HA)** upgrade mechanisms.
- Developed a real-time upgrade workaround system enabling interactive FMC-FTD sessions using REST APIs, **reducing customer support cases by 50%**.
- Built a **NetSec ORBIT MCP-based extension** for Windsurf and VS Code to auto-triage bugs via CDET MCP, identify impacted branches, and perform system-level root cause analysis, **reducing bug turnaround time by 60%**.
- Engineered an optimized upgrade mechanism using the SquashFS file system, achieving an **80% reduction in storage usage** and a **75% improvement in upgrade time** by adopting an install-based approach.
- Integrated core networking protocols including **SSH, DNS, DNSSEC, DHCP, SCP, and SFTP**, enabling secure data transfer and authenticated name resolution while **reducing application latency**.
- Enforced a **centralized logging system** using socket programming to collect, parse, and process real-time logs from LINA and other services within a scalable microservices architecture.
- Optimized the FTD patch upgrade process by enabling **direct patch installation**, bypassing intermediate MR upgrades and resulting in a **40% reduction in upgrade and revert time**.

Technical Undergraduate Intern, Cisco Systems, Bengaluru, India May 2021 – July 2021

- Automated dynamic scaling of **Web Security Appliance (WSA)** instances on AWS based on real-time traffic and load metrics.
- Built a CLI to aggregate and present data from multiple WSA appliances using REST APIs implemented with the **Boto3** library.

EDUCATION

National Institute Of Technology Karnataka, Surathkal July 2018 - May 2022

B.Tech in Electronics and Communication Engineering and Minor in Computer Science

PROJECTS

Auto Bug Classifier 2025

- Designed an intelligent tool with **MCP servers merged with CDET and Confluence**, enabling automatic bug detail retrieval and correlation with the codebase for root cause analysis, while leveraging the Autopsy server for efficient **core file debugging**.

Encrypted Service [Link](#)

- Developed a secure web communication application incorporating **steganography** for data hiding and **SHA-256** for encryption.
- Used **Diffie-Hellman** algorithm for secure key exchange, paired with **256-bit encryption** for secure messaging.

ACHIEVEMENTS

- Published a **research paper** on **Evolution of LiverNet 2.x**, a lightweight CNN for liver cancer classification. [LiverNet 2.x](#)
- Got **third place in company-wide hackathon**; the solution was applied into the product and **reduced TAC cases by 60%**.
- Received the **Engineering Pioneers in Innovation and Creativity (EPIC)** award for space optimization.
- **Codeforces Expert** with a **current rating of 1761**; solved **260+ problems** across LeetCode. [Codeforces](#) [LeetCode](#)
- Achieved a **6-star rating** in Problem Solving on HackerRank. [HackerRank](#)